

Online Supplement: Adjustment Margins and Tax–Benefit Cushioning of Earnings Losses in the United Kingdom

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S1 Public product–destination benchmark: design

To test whether post-policy US exports move differently from other UK export destinations, the main paper uses a monthly HMRC OTS panel from January 2018 to May 2026 for the United States, Canada, Japan and Australia (HM Revenue and Customs, 2026). Detailed HS8 rows are aggregated to HS4 and balanced across products, destinations and months, with absent flows set to zero. HMRC’s separate two-digit aggregate rows are excluded. For product p and month t , the outcome is

$$g_{pt} = \log(1 + V_{p,\text{US},t}) - \frac{1}{3} \sum_{d \in \{\text{CA}, \text{JP}, \text{AU}\}} \log(1 + V_{pdt}).$$

The destination gap is regressed on a post-May-2025 indicator, HS4 product fixed effects, month-of-year effects and a linear trend. April 2025 is omitted as an announcement and anticipation month. The primary WLS weights are products’ mean 2022–24 US export values, capped at the 99th percentile; standard errors are clustered by HS4. Uncapped value weights, equal product weights, individual destinations, later sample starts and pseudo-policy dates in May 2019–23 are reported as specification checks.

This design controls for product shocks common across destinations but not US-specific demand, exchange rates, shipping, inventories, composition or other coincident policies. The high-tariff comparison groups HS chapters 72, 73 and 87 as steel and automotive products; that broad mapping is a falsification check, not a complete product-level statutory tariff schedule. The panel is therefore not used to estimate the demand or incidence parameters of the main paper’s accounting bridge. It tests whether the imposed bridge is consistent with destination-differential outturn patterns.

S2 HMRC destination-panel estimates

Across 1194 HS4 products, the capped value-weighted specification estimates a -31.9 per cent change in exports to the US relative to the same products sent to the three comparison destinations (95 per cent interval -44.8 to -16.0 per cent, $p < 0.001$). Results have similar signs against each destination, and May 2019–23 pseudo-policy estimates are individually insignificant. This is not robust evidence of a common product effect: equal product weights give -4.9 per cent ($p = 0.293$), and restricting the sample to 2023 onward gives -17.5 per cent ($p = 0.080$). The estimated differential trend is itself borderline ($p = 0.057$). Most importantly, broad steel and auto chapters fall -20.0 per cent ($p = 0.290$), compared with -32.9 per cent for other products ($p = 0.001$): the tariff-intensity ordering fails. Figure S1 displays the monthly gap. The panel supports a post-policy decline concentrated in large US export products, but it cannot attribute that decline to tariffs or validate the tariff-rate mapping.

The machine-readable specification, clustered standard errors, individual destination checks, sample-start sensitivities, tariff-intensity groups and May 2019–23 pseudo-policy estimates are stored in the event-study JSON artifact; the plotted observations are stored in its companion monthly CSV. Pre-policy value weights are capped at the 99th percentile, giving an effective product count of 107 despite 1194 included HS4 products. Uncapped and equal-weight estimates disclose the remaining concentration sensitivity.

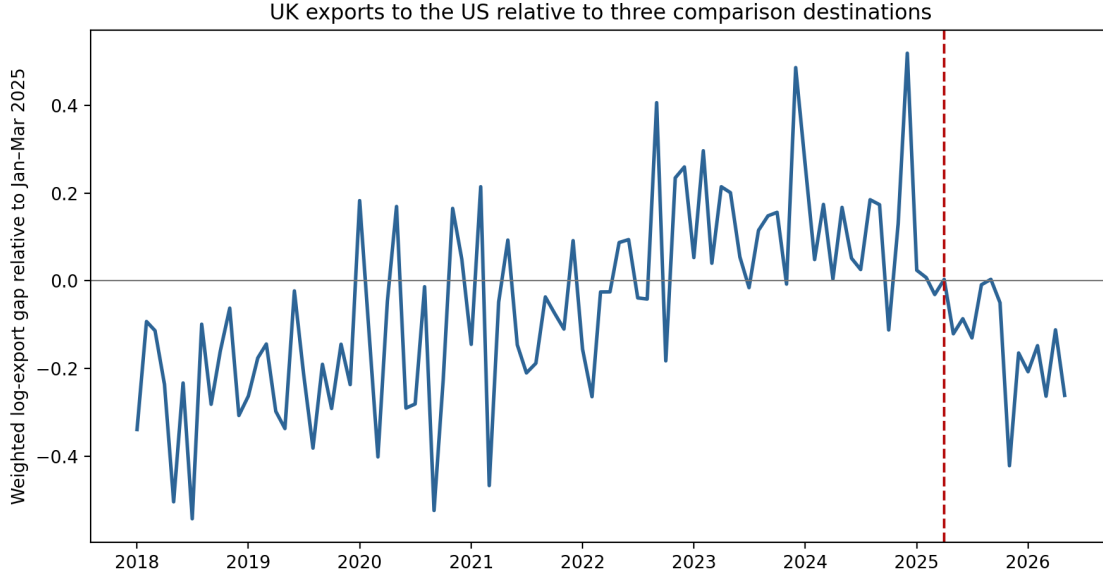


Figure S1: Monthly capped-value-weighted gap between UK HS4 exports to the United States and mean exports of the same products to Canada, Japan and Australia, normalised to January–March 2025.

S3 Longitudinal LFS imputation benchmark: design

The uniform within-division assignment in the central displacement family is transparent but suppresses observable worker heterogeneity. UK Data Service Study 9490, the Labour Force Survey Five-Quarter Longitudinal Dataset covering July 2024–September 2025, provides a donor benchmark. The donor population comprises respondents employed at wave 1. Job exit is observed at wave 5; log wage change is defined only for respondents employed with positive main-job pay at both endpoints. Negative UKDS sentinel values are missing, not zero. Public 2024 Nomis BRES employee totals rescale the manufacturing SIC composition before direct transition targets are computed. ([Office for National Statistics, 2025](#))

The primary imputation maps SIC-division–sex–age cells to employed FRS receivers, shrinking sparse cells towards sector and manufacturing means. After matching, predicted exit risks are rescaled so their FRS-weighted mean equals the directly measured, BRES-composition-adjusted LFS manufacturing exit rate. Mean predicted log wage changes are centred on the corresponding LFS target. An income-tercile estimator supplies a transparent sensitivity specification.

A regularised quantile/random-forest benchmark follows the calibrated imputation architecture in PolicyEngine’s `young-worker-nics` project. It estimates model shape on all wave-1 employed adult donors using age, sex and annualised employment income. The binary outcome uses the random-forest event probability; the continuous outcome uses the QRF conditional mean. Forests use 500 trees and a minimum leaf size of 20; wage outcomes are winsorised at the donor 1st and 99th percentiles. Both receiver predictions are then calibrated to exactly the same manufacturing LFS targets as the cell estimator. This is a predictive baseline-transition benchmark, not an estimate of tariff-induced exits or wage changes. It is therefore reported as a robustness diagnostic and does not replace the declared central assignment rule.

In a downstream allocation sensitivity, each receiver score r_i is converted to a within-division

displacement probability $p_i = \min(1, k_j r_i)$, where k_j is solved numerically so

$$\frac{\sum_{i \in j} p_i E_i w_i}{\sum_{i \in j} E_i w_i} = s_j.$$

Thus cells, earnings bands and QRF change who is exposed while preserving each division’s expected employee-wage-bill loss. Missing scores use the division mean. Common assignments then isolate sensitivity to incidence shape; they do not give the baseline transition model a causal tariff interpretation.

S4 Longitudinal worker benchmark: results

The five-quarter LFS contains 177 wave-1 manufacturing employees after the required population filters, compared with 1102 employed manufacturing FRS receivers. Its BRES-composition-adjusted manufacturing targets are an 8.96 per cent wave-1-to-wave-5 job-exit rate and 5.87 per cent mean log wage growth among valid endpoint-pay observations. Table S1 compares receiver distributions. Calibration deliberately makes the aggregate exit means identical, so the comparison tests incidence shape rather than level. The QRF has a much wider risk distribution and agrees weakly with the primary cells: person-level correlations are only 0.06 for exit risk and 0.08 for predicted wage change. The longitudinal evidence therefore does not support replacing transparent cells with forest-generated tariff incidence. It instead quantifies model uncertainty that the central uniform assignment omits. Regularisation does not eliminate all forest tails: the maximum calibrated exit risk is 62.6 per cent and the predicted log-wage-change range is -45.7 to 72.0 per cent. These are benchmark predictions, not plausible confidence bounds.

Table S1: Calibrated FRS job-exit-risk distributions

Estimator	Weighted mean	P10	Median	P90
Calibrated SIC–sex–age cells	8.96	7.15	8.42	11.88
Calibrated income terciles	8.96	3.88	8.91	13.77
Regularised QRF benchmark	8.96	1.22	5.42	19.62

Notes: Entries are percentages among matched employed manufacturing FRS receivers. All means equal the same directly measured LFS target by construction. Quantiles are survey-weighted. QRF denotes the regularised quantile/random-forest benchmark described in Section S3; it predicts baseline transitions, not causal tariff effects.

Fiscal sensitivity to worker-risk shape. Table S2 uses each calibrated risk model to vary selection within SIC divisions while holding every division’s expected weighted wage-bill loss at the original shock. The cell model nearly reproduces uniform assignment. Across all three LFS-shaped models, mean cushioning ranges from 36.0 to 37.4 per cent, Exchequer cost from £384 million to £425 million, and measured poverty change from 0.024 to 0.028 percentage points. This model spread is smaller than the assignment SD within any specification. Weighted displaced headcount changes because calibration preserves the wage bill, not a headcount quota. The QRF mean gross loss is £847 million rather than the £886 million expectation, but its £397 million assignment SD implies a numerical MC SE of about £40 million, so this difference is not evidence of a different aggregate target.

Table S2: Full-schedule fiscal sensitivity to within-SIC worker selection

Selection model	Workers (000s)	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
Uniform central assignment	17.2 $\pm$ 6.9	882 $\pm$ 418	387 $\pm$ 213	34.1 $\pm$ 5.2	0.032 $\pm$ 0.022
LFS calibrated cells	17.2 $\pm$ 6.9	881 $\pm$ 412	412 $\pm$ 223	36.9 $\pm$ 5.0	0.028 $\pm$ 0.019
LFS income terciles	15.2 $\pm$ 6.5	887 $\pm$ 467	425 $\pm$ 258	37.4 $\pm$ 6.0	0.025 $\pm$ 0.017
LFS regularised QRF	18.1 $\pm$ 5.9	847 $\pm$ 397	384 $\pm$ 212	36.0 $\pm$ 5.7	0.024 $\pm$ 0.016

Notes: Means $\pm$ assignment SD over 100 common assignments. Within each SIC division, individual probabilities are rescaled so the earnings-weighted mean equals the declared sector shock. Poverty is relative BHC against the frozen baseline line. LFS risks are predictive baseline-transition scores, not tariff-specific causal probabilities.

S5 EPD counterfactual

On paired seeds, full minus EPD displacement differs by 3.7 ± 2.8 thousand workers and $\pounds 90 \pm 98$ million of Exchequer cost. The contrast is driven primarily by the lower auto tariff and the exemption of pharmaceuticals in the EPD schedule. It should be interpreted as the fiscal value of changing the declared sector shock while holding the household model and assignment draws fixed. It does not include quota allocation, above-quota trade, price responses, NHS pricing concessions, or other general-equilibrium effects. Figure S2 summarises the paired comparison.

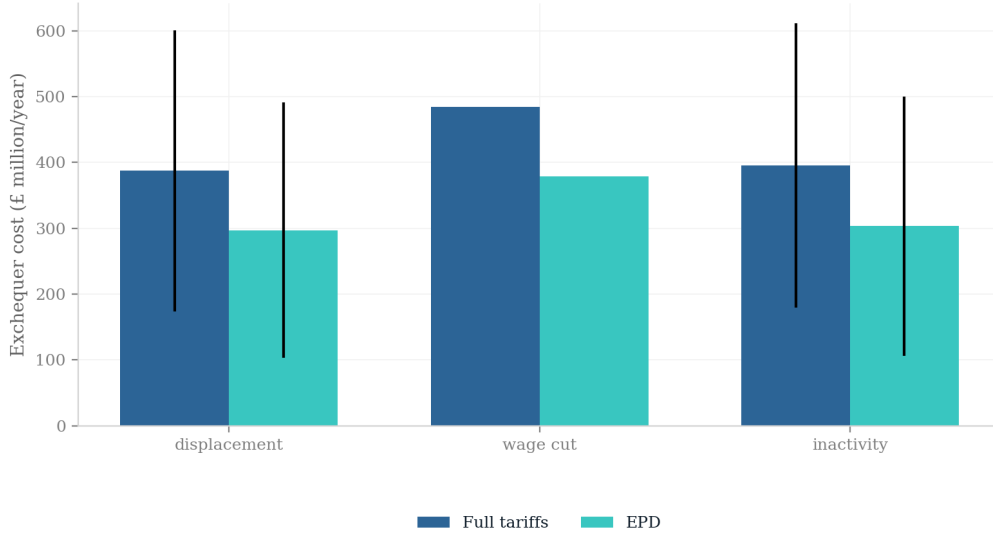


Figure S2: Full-schedule and EPD scenario comparison. Error bars are assignment standard deviations.

S6 Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{0.4, 1, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

S7 Exploratory mixed-adjustment surface

The exploratory scenario surface separates the aggregate export-demand calibration from the assumed composition of labour-market adjustment. Each cell uses the full tariff schedule and five common assignments. The wage-cut endpoint is stable: cushioning is 40.6–41.3 per cent over the four calibrations. Moving towards displacement generally lowers cushioning, but the small-shock cells are especially noisy because a few high-weight FRS records dominate five Bernoulli assignments. These cells are stress scenarios, not estimates or confidence regions. The qualitative result is that fiscal cost scales mainly with the imposed gross shock, while the adjustment mix changes the share absorbed by taxes and transfers. Its machine-readable cells and draws are in `results/scenario_testing.json` and `results/scenario_testing.csv`.

S8 Reallocation sensitivity

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £264 million, Exchequer cost of $£149 \pm 131$ million, cushioning of 39.7 ± 3.4 per cent, and zero measured BHC-poverty change. With a three-month lag, gross loss rises to £432 million and the Exchequer cost to $£245 \pm 216$ million. The 14 per cent penalty gives gross loss of £131 million and Exchequer cost of $£75 \pm 64$ million. These pound levels are not commensurable with the economy-wide wage-cut scenario because only selected movers’ penalty is removed.

S9 Observed-outturn stress scenario

The descriptive May 2025–February 2026 export window falls 12.7 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates, front-running, rerouting and other demand conditions.

The downside displacement scenario affects 17.3 thousand weighted workers on average, is cushioned by 34.0 ± 5.0 per cent and costs the Exchequer $£394 \pm 218$ million. The wage-cut scenario is cushioned by 44.1 per cent and costs £517 million. This is an observed-outturn downside stress case, not a tariff-caused or preferred estimate; the validation artifact also reports signed net exposure before expanding divisions are clipped.

S10 Demographic diagnostic

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. Selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The demographic artifact is therefore a diagnostic rather than an inferential table.

S11 Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound

and is not multiplied by local-employment multipliers. In the exploratory ten-assignment microsimulation, displacement affects 29,700 weighted workers on average, costs the Exchequer $\pounds 680 \pm 310$ million and raises relative BHC poverty by 0.028 percentage points. The wage-cut variant costs $\pounds 960$ million and leaves measured relative poverty unchanged. These figures are accounting sensitivities, not all-channel causal estimates.

S12 Constituency machinery

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. Its outputs are averaged over 20 paired worker-assignment draws (seeds 0–19), with each household’s disposable-income change divided by household size before constituency means are constructed; the CSV artifact reports both the draw mean and the assignment-draw standard deviation. Because the constituency weights contain no observed local industry mix and the SIC imputation is held fixed, the outputs are synthetic demographic and income projections, not constituency exposure estimates. No map or seat-level figures are reported in the main paper; the machinery is documented here so that future work can populate it with observed local industry data.

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